

LAPORAN PRAKTIKUM
OTOMASI INDUSTRI
UNIT 2



Oleh :

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Kelas : TRE B2
Hari, Tanggal : Selasa, 27 Februari 2024
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PROGRAM STUDI TEKNOLOGI REKAYASA ELEKTRO
DEPARTEMEN TEKNIK ELEKTRO DAN INFORMATIKA
SEKOLAH VOKASI
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A. Tujuan

Setelah melaksanakan kegiatan pertemuan unit 2, mahasiswa diharapkan mampu:

1. Memahami jenis-jenis karakteristik jalur masukan PLC
2. Memahami jenis-jenis karakteristik jalur keluaran PLC
3. Memahami karakteristik kelistrikan piranti masukan dan keluaran
4. Merancang sambungan antara PLC dengan piranti masukan dan piranti keluaran

B. Alat dan Bahan

Alat dan bahan yang digunakan untuk melaksanakan praktikum unit ini adalah:

1. Komputer
2. Koneksi internet
3. Alat tulis

C. Langkah Kerja

1. Buka tautan ugm.id/otomasib. Buka file PLC P1E
2. Identifikasi rangkaian internal jalur masukan PLC anda
3. Gambar hasil langkah nomor 2.
4. Identifikasi rangkaian internal jalur keluaran PLC anda
5. Gambar hasil langkah nomor 4.

D. Data Sementara

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Tantangan SOAL

1. Jalur keluaran dari PLC yang anda gunakan, memiliki piranti pensaklaran jenis apa?
2. Gunakan program penjelajah internet untuk mendapatkan:
 - 20 buah sensor yang dapat dibaca kondisinya oleh modul masukan diskrit PLC
 - 20 buah aktuator yang dapat dikendalikan oleh modul keluaran diskrit PLC
3. Lengkapi sensor dan aktuator dengan lembar datanya (Datasheet)

Lembar kerja unit 2, piranti masukan dan keluaran Programmable Logic Controller. Hasil identifikasi rangkaian internal jalur masukan PLC anda

Gambar hasil Langkah nomor 2 dan identifikasi rangkaian internal jalur keluaran PLC anda

Gambar hasil Langkah nomor 4

Jawaban

1. PLC CP1E type E20SDR-A memiliki piranti pensaklaran yang berjenis relay serta memiliki program kapasitas yang berjumlah 2K Langkah dan memori kapasitas data berjumlah 2K kata
2. Sensor yang dapat dibaca kondisinya oleh modul masukan diantaranya:
 - Sensor photoelectric
 - Sensor proximity
 - Sensor pengukuran
 - Sensor thermocouple
 - Encoder
 - Sensor ultrasonic
 - Sensor tekanan
 - Sensor kecepatan
 - Sensor level
 - Sensor gas
 - Sensor cahaya
 - Sensor getaran
 - Sensor kelembapan
 - Sensor pH
 - Sensor arus
 - Sensor tegangan
 - Sensor suhu
 - Sensor kedalaman
 - Sensor magnetic
 - Sensor kapasitif

Actuator yang dapat dikendalikan oleh modul keluaran diantaranya:

- Motor DC
- Motor AC
- Solenoid Valve
- Relay
- Lampu
- Heater
- Fan
- Pneumatic cylinder
- Hydraulic cylinder
- Electromagnet
- Buzzer
- Selenoid
- Kontaktor
- Servo motor
- Stepper motor
- Actuator linear
- Actuator rotary
- Valve
- Pump
- Electropneumatic valve

Item	Specification		
Input type	High-speed counter inputs or Normal inputs	High-speed counter inputs, Interrupt input, quick-response inputs, or Normal inputs	Normal inputs
Input bits	CIO 0.00 to CIO 0.01	CIO 0.02 to CIO 0.07 #1	CIO 0.08 to CIO 0.11, CIO 1.00 to CIO 1.11 and CIO 2.00 to CIO 2.11 #1
Input voltage	24 VDC, +10%, -15%		
Applicable sensors	2-wire and 3-wire sensors		
Input impedance	3.3 k Ω	3.3 k Ω	4.8 k Ω
Input current	7.5 mA typical	7.5 mA typical	5 mA typical
ON voltage/current	3 mA min. at 17.0 VDC max.	3 mA min. at 17.0 VDC max.	3 mA min. at 14.4 VDC min.
OFF voltage/current	1 mA max. at 5.0 VDC max.	1 mA max. at 5.0 VDC max.	1 mA max. at 5.0 VDC max.
ON response time #2	E...[S]-type CPU Unit: 50 μ s min. NNA...[S]-type CPU Unit: 2.5 μ s min.	50 μ s max.	1 ms max.
OFF response time #2	E...[S]-type CPU Unit: 50 μ s min. NNA...[S]-type CPU Unit: 2.5 μ s min.	50 μ s max.	1 ms max.
Circuit configuration	E...[S]-type CPU Unit		NNA...[S]-type CPU Unit
	Input 0.00 to 0.07		Input 0.00 to 0.07
	Input 0.08 to 0.11, 1.00 to 1.11		Input 0.02 to 0.07
			Inputs CIO 0.08 to CIO 0.11, CIO 1.00 to CIO 1.11 and CIO 2.00 to CIO 2.11

Identifikasi rangkaian internal jalur masukan PLC CP1E-E20SDR sebagai berikut

Jawaban:

Input

- Terdiri dari tiga channel: channel 0 dengan 12 input individu, saluran 90 dan saluran 91 masing-masing dengan satu input analog resolusi 6000
- Input dapat diperluas

Identifikasi rangkaian internal jalur keluaran PLC CP1E-E20SDR sebagai berikut

Jawaban:

Output

- Memiliki dua saluran: saluran 100 termasuk output diskrit dan saluran 190 berisi satu output analog resolusi 2000
- Output dapat diperluas untuk mereproduksi output dan input

1. Karakteristik kelistrikan keluaran PLC CP1E-E20SDR-A

Memiliki tipe keluaran berupa relay dengan spesifikasi sebagai berikut.

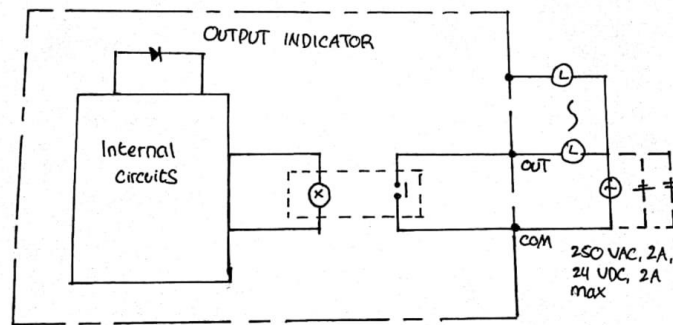
Kapasitas switching maksimum:

1. 250VAC/2A dengan cosphi adalah 1
2. 24VDC/2A

Durabilitas

1. Beban resistif dapat beroperasi 200.000 kali dengan 24VDC
2. Beban induktif dapat beroperasi 70.000 kali dengan 250 VAC cosphi 0.4

Rangkaian Internal Jalur Keluaran



Karakteristik kelistrikan masukan PLC CP1E-E14DR-A

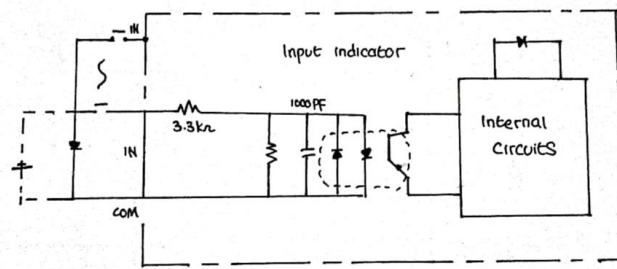
CIO 0.00 hingga CIO 0.01

1. Memiliki input tipe High-speed counter inputs or Normal
2. Input tegangan 24 VDC +10%, -15%
3. Sensor yang dapat diaplikasikan merupakan sensor 2 wire dan 3 wire
4. Impedansi input 3,3 K Ω
5. Nilai arus input 7,5 mA
6. Arus dan tegangan ketika beroperasi memiliki nilai minimum 3 mA dan 17 VDC
7. Arus dan tegangan ketika tidak beroperasi memiliki nilai maksimum 1 mA dan 5 VDC

CIO 0.02 hingga 0.07

1. High-speed counter inputs, interrupt, input, quick-response inputs, or Normal Inputs
2. Input tegangan 24 VDC +10%, -15%
3. Sensor yang dapat diaplikasikan merupakan sensor 2 wire dan 3 wire
4. Impedansi input 3,3 K Ω
5. Nilai arus input 7,5 mA
6. Arus dan tegangan ketika beroperasi memiliki nilai minimum 3 mA dan 17 VDC
7. Arus dan tegangan ketika tidak beroperasi memiliki nilai maksimum 1 mA dan 5 VDC

Rangkaian Internal Jalur Masukan



E. Sensor yang dapat dibaca kondisinya oleh modul masukan diantaranya:

- Sensor photoelectric
- Sensor proximity
- Sensor pengukuran
- Sensor thermocouple
- Encoder
- Sensor ultrasonic
- Sensor tekanan
- Sensor kecepatan
- Sensor level
- Sensor gas
- Sensor cahaya
- Sensor getaran
- Sensor kelembapan
- Sensor pH
- Sensor arus
- Sensor tegangan
- Sensor suhu
- Sensor kedalaman
- Sensor magnetic
- Sensor kapasitif

Actuator yang dapat dikendalikan oleh modul keluaran diantaranya:

- Motor DC
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- Solenoid Valve
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- Electromagnet
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- Selenoid
- Kontaktor
- Servo motor
- Stepper motor
- Actuator linear
- Actuator rotary
- Valve
- Pump
- Electropneumatic valve

F. Lengkapi sensor dan actuator dengan lembar datanya
 Karakteristik kelistrikan PLC Schneider TM3DM24R



- a. Modul Masukan
 - 1) Memiliki 16 jalur input
 - 2) Memiliki tipe input *sinking* dan *sourcing*
 - 3) Memiliki tegangan masukan 24 VDC
 - 4) Memiliki tegangan untuk komunikasi 5 VDC
 - 5) Modul mengonsumsi arus 7 mA
- b. Modul keluaran
 - 1) Memiliki 8 jalur output
 - 2) Memiliki tipe output relay dengan kondisi NO
 - 3) Memiliki metode output berupa gerbang NPN
 - 4) Memiliki tegangan rating 24 VDC atau 240 VAC
 - 5) Modul bekerja dengan tegangan maksimum 30 vdc
 - 6) Arus maksimum yang dapat bekerja adalah 2 A
 - 7) Modul akan menyebabkan drop tegangan maksimum 2,5 V

G. Karakteristik kelistrikan PLC Mitsubishi FX2N-2LC



Modul masukan

1. Memiliki 4 jalur input
2. Memiliki tegangan rating 24 VDC
3. Memiliki tegangan rating untuk komunikasi 5VDC
4. Modul akan mengonsumsi arus 55 mA ketika tegangan 24 VDC dan 70 mA ketika tegangan 5 VDC

Modul keluaran

1. Memiliki 2 jalur output
2. Memiliki tipe output berupa relay
3. Memiliki metode output berupa gebrang NPN
4. Memiliki tegangan rating 5 hingga 24 VDC
5. Modul bekerja dengan tegangan maksimum 30 VDC
6. Modul bekerja dengan arus maksimum 100 mA
7. Modul akan menyebabkan drop tegangan maksimum 2,5 V

H. Jenis Sensor

1. Rotary Encoder



Model		E6CP-AG3C	E6CP-AG5C	E6CP-AG5C-C
Power supply voltage		5 VDC -5% to 12 VDC +10%, ripple (p-p): 5% max.		12 VDC -10% to 24 VDC +15%, ripple (p-p): 5% max.
Current consumption *1		90 mA max.	70 mA max.	
Resolution (rotations)		256 (8-bit)		
Output code		Gray code		
Output configuration		Open-collector output		
Output capacity		Applied voltage: 28 VDC max. Sink current: 16 mA max. Residual voltage: 0.4 V max. (at sink current of 16 mA)		
Maximum response frequency *2		5 kHz		
Logic		Negative logic (high = 0, low = 1)		
Accuracy		±1° max.		
Direction of rotation		Output code incremented by CW (as viewed from the end of the shaft)		
Rise and fall times of output		1 μs max. (Control output voltage: 16 V, Load resistance: 1 kΩ, Output cable: 2 m max.)		
Starting torque		0.98 mN·m max.		
Moment of inertia		1 × 10 ⁻⁶ kg·m ² max.		
Shaft loading	Radial	29.4 N		
	Thrust	19.6 N		
Maximum permissible speed		1,000 r/min		
Ambient temperature range		Operating: -10 to 55°C (with no icing), Storage: -25 to 85°C (with no icing)		
Ambient humidity range		Operating/Storage: 35% to 85% (with no condensation)		
Insulation resistance		200 MΩ min. (at 500 VDC) between current-carrying parts and case		
Dielectric strength		500 VAC, 50/60 Hz for 1 min between current-carrying parts and case		
Vibration resistance		Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 hours each in X, Y, and Z directions		
Shock resistance		Destruction: 1,000 m/s ² 3 times each in X, Y, and Z directions		
Degree of protection*3		IEC 60529 IP50		
Connection method		Pre-wired Models (Standard cable length: 2 m)		Connector Models (Standard cable length: 2 m)
Material		Case: ABS, Main unit: PPS, Shaft: SUS416, Mounting Bracket: Galvanized iron		
Weight (packed state)		Approx. 200 g		
Accessories		Coupling (excluding Connector Models), Servo Mounting Bracket, Hexagonal wrench (excluding Connector Models), Instruction manual		

2. Ultrasonic Sensor



Model	E4C-DS30	E4C-DS30L	E4C-DS80	E4C-DS80L	E4C-DS100
Measurement range	60 to 275 mm		85 to 735 mm		110 to 910 mm
Standard sensing object	100 × 100 mm SUS flat plate				
Near distance dead band	0 to 50 mm		0 to 70 mm		0 to 90 mm
Ultrasonic oscillation frequency	Approx. 390 kHz		Approx. 255 kHz		
Response speed *	30 ms		100 ms		125 ms
Ambient temperature range	Operating: .25 to 70°C, Storage: .40 to 85°C (with no icing or condensation)				
Ambient humidity range	Operating and storage: 35% to 85% (with no condensation)				
Insulation resistance	50 MΩ min. (at 500 VDC)				
Dielectric strength	1,000 VAC, 50/60 Hz for 1 min				
Vibration resistance	10 to 55 Hz, 1.5-mm double amplitude, 2 hours each in X, Y, and Z directions				
Shock resistance	500 m/s ² , 3 times each in X, Y and Z directions				
Enclosure rating	IP65				
Indicator	(Yellow) Lit: Sensor within sensing range (Green) Lit: Power indicator				(Yellow) Lit: Sensor within sensing range
Weight	Approx. 150 g				Approx. 170 g
Materials	Case: Nickel-plated brass, Oscillator surface: Glass epoxy resin and polyurethane				
Accessories	Instruction Manual, XS2F-D523-D80-A (Cable length: 2 m), XN2A-1430				

3. High Temperature Cleaning Sensor



Size Shielded Model		M12	M18	M30
		Shielded		
		E2EH-X3D[]	E2EH-X7D[]	E2EH-X12D[]
Sensing distance		3 mm	7 mm	12 mm
Set distance *1		0 to 2.4 mm	0 to 5.6 mm	0 to 9.6 mm
Differential travel		15% max. of sensing distance		
Detectable object		Ferrous metal (The sensing distance decreases with non-ferrous metal. Refer to Datasheet.)		
Standard sensing object		Iron, 12 × 12 × 1 mm	Iron 21 × 21 × 1 mm	Iron 36 × 36 × 1 mm
Response frequency *2		500 Hz	300 Hz	100 Hz
Power supply voltage (operating voltage range)		12 to 24 VDC, ripple (p-p): 10% max. (10 to 32 VDC, however, 24 VDC max. at temperatures over 100 °C)		
Leakage current		0.8 mA max.		
Control output	Load current	3 to 100 mA (however, 3 to 50 mA at 100 to 110 °C)		
	Residual voltage *3	Polarity Models: 3 V max. No polarity Models: E2EH-X[]D[]-T : (5 V max. *3 (Load current: 100 mA, Cable length 2 m))		
Indicators		D1 Models: Operation indicator (red), Setting indicator (yellow) D2 Models: Operation indicator (yellow)		
Operation mode (with sensing object approaching)		D1 Models: NO D2 Models: NC Refer to Datasheet.		
Protection circuits		Surge suppressor, Load short-circuit protection		
Ambient temperature range		Operating: 0 to 100 °C (0 to 110 °C 1,000 h) *4 Storage: - 25 to 70 ° (with no icing or condensation)		
Ambient humidity range		35% to 95%		
Temperature influence		± 10% max. of sensing distance at 23 °C in the temperature range of 0 to 70 °C. ± 15% max. of sensing distance at 23 °C in the temperature range of 70 to 100 °C. - 15% to +20% of sensing distance at 23 °C in the temperature range of 100 to 110 °C.		
Voltage influence		± 10% max. of sensing distance at rated voltage in the 15% rated voltage range.		
Insulation resistance		50 MΩ min. (at 500 VDC) between current-carrying parts and case		
Dielectric strength		1,000 VAC, 50/60 Hz for 1 min between current-carrying parts and case		
Vibration resistance		Destruction: 10 to 55 Hz 1.5-mm double amplitude for 2 hours each in X, Y and Z directions		
Shock resistance		Destruction: 1,000 m/s ² , 10 times each in X, Y and Z directions		
Degree of protection		IEC IP67, DIN 40050-9 IP69K *5		
Connection method		Pre-wired Models (Standard cable length 2 m), Connector Models		
Weight (packed state)	Pre-wired Models	Approx. 80 g	Approx. 145 g	Approx. 220 g
	Connector Models	Approx. 30 g	Approx. 55 g	Approx. 125 g
Materials	Case, clamping nut	Stainless steel (SUS316L)		
	Sensing surface	PBT		
	Cable	Heat-resistant PVC cable (Pre-wired model)		
Accessories		Instruction manual		

4. Aluminium Detecting Proximity Sensor



Model		E2EY-X4C1	E2EY-X8C1
Sensing distance		4 mm $\pm$ 10%	8 mm $\pm$ 10%
Set distance		0 to 2.8 mm	0 to 5.6 mm
Differential travel		20% max. of sensing distance	
Detectable object		Non-ferrous metal (Does not detect ferrous metal.)	
Standard sensing object		Aluminum: 18 × 18 × 1 mm	Aluminum: 30 × 30 × 1 mm
Response frequency *		70 Hz	
Power supply voltage (operating voltage range)		12 to 24 VDC (10 to 30 VDC), ripple (p-p): 10% max.	
Current consumption		20 mA max.	
Control output	Load current	NPN open-collector output, 100 mA max. (at 30 VDC)	
	Residual voltage	2 V max. (Load current: 100 mA, Cable length: 2 m)	
Indicators		Detection indicator (red)	
Operation mode (with sensing object approaching)		Load ON: NO (Refer to Datasheet.)	
Protection circuits		Reverse polarity protection, Load short-circuit protection, Surge suppressor	
Ambient temperature range		Operating/Storage: -10 to 55°C (with no icing or condensation)	
Ambient humidity range		Operating/Storage: 35% to 95% (with no condensation)	
Temperature influence		$\pm$ 20% max. of sensing distance at 23°C in the temperature range of -10 to 55°C	
Voltage influence		$\pm$ 2.5% max. of sensing distance at rated voltage in rated voltage $\pm$ 15% range	
Insulation resistance		50 M Ω min. (at 500 VDC) between current-carrying parts and case	
Dielectric strength		1,000 VAC, 50/60 Hz for 1 minute between current-carrying parts and case	
Vibration resistance		Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 hours each in X, Y, and Z directions	
Shock resistance		Destruction: 1,000 m/s ² 10 times each in X, Y, and Z directions	
Degree of protection		IEC 60529 IP67, in-house standards: oil-resistant	
Connection method		Pre-wired Models (Standard cable length: 2 m)	
Weight (packed state)		Approx. 140 g	Approx. 190 g
Materials	Case	Nickel-plated brass	
	Sensing surface	Heat-resistant ABS	
	Clamping nuts	Nickel-plated brass	
	Toothed washer	Zinc-plated iron	
Accessories		---	

5. Photosensor Through Beam



Models		EE-SPW321, EE-SPW421	EE-SPW321-A, EE-SPW421-A
Sensing distance		300 mm *1	
Sensing object		Opaque: 2 mm dia. min. *2	
Directional angle		10° to 40°	
Light source		GaAs infrared LED (pulse lighting) with a peak wavelength of 940 nm	
Indicator		Light indicator (Red LEDs, one each on Sensor and Amplifier)	
Supply voltage		12 to 24 VDC $\pm$ 10%, ripple (p-p): 5% max.	
Current consumption		Average: 30 mA max.	
Control output		NPN open collector, Load power supply voltage: 12 to 24 VDC, Load current: 100 mA max., OFF current: 0.5 mA max. Residual voltage: 1 V max (at a 100-mA load current)	
Response time		1 ms max. for both detection and reset	
Ambient illumination		3,000 lx max. (incandescent light); 10,000 lx max. (sunlight) on the receiver	
Ambient temperature range		-20 to +55°C	
Ambient humidity range		5% to 85%	
Vibration resistance		Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 h each in X, Y, and Z directions	
Shock resistance		500 m/s ²	
Degree of protection		IEC IP64	
Connecting method		Pre-wired (standard cable length: 2 m)	
Cable length from emitter (receiver) to amplifier		0.5 m	1 m
Weight (Packaged)		76 g	
Material	Case	ABS resin	
	Lens	Acrylate resin	
Accessories		Slits: One sheet with 16 total slits (0.5 × 3 mm, 1 × 3 mm, 3 × 0.5 mm, 3 × 1 mm; four each) included Sems screws with spring washers and flat washers: Six M2.6 × 12 Instruction Manual	

6. Photosensor Limited-Reflective



Models		EE-SPY311, EE-SPY411, EE-SPY312, EE-SPY412
Sensing distance		2 to 5 mm (Reflection factor: 90%; white paper 15 × 15 mm)
Minimum sensing object		Pure copper wire (0.05 mm dia.)
Distance to background *1		20 mm max. (glass with aluminum deposition)
Differential distance		0.2 mm (with a sensing distance of 3 mm, horizontally)
Light source		GaAs infrared LED with a peak wavelength of 940 nm
Indicator *2		Light indicator (red)
Supply voltage		5 to 24 VDC ±10%, ripple (p-p): 5% max.
Current consumption		Average: 15 mA max., Peak: 50 mA max.
Control output		NPN voltage output: Load power supply voltage: 5 to 24 VDC Load current: 80 mA max. OFF current: 0.5 mA max. 80 mA load current with a residual voltage of 1.0 V max. 10 mA load current with a residual voltage of 0.4 V max.
Response frequency *3		100 Hz min.
Ambient illumination		3,000 lx max. with incandescent light or sunlight on the surface of the receiver
Ambient temperature range		Operating: - 10 to +55 ° C Storage: - 25 to +65 ° C
Ambient humidity range		Operating: 5% to 85% Storage: 5% to 95%
Vibration resistance		Destruction: 10 to 50 Hz, 1.5-mm double amplitude for 2 h each in X, Y, and Z directions
Shock resistance		Destruction: 500m/s ² for 3 times each in X, Y, and Z directions
Degree of protection		IEC IP50
Connecting method		Special connector (soldering not possible)
Weight		Approx. 2.6 g
Material	Case	Polycarbonate
	Holder	Polybutylene phthalate (PBT)

7. Welding Proximity Sensor



Type		Double distance model/ Spatter-resistant Double distance model			Single distance model/ Spatter-resistant Single distance model		
Size		M12	M18	M30	M12	M18	M30
Model		E2EW- (Q)X3□12	E2EW- (Q)X7□18	E2EW- (Q)X12□30	E2EW- (Q)X2□12	E2EW- (Q)X5□18	E2EW- (Q)X10□30
Sensing distance		3 mm ±10%	7 mm ±10%	12 mm ±10%	2 mm ±10%	5 mm ±10%	10 mm ±10%
Setting distance		0 to 2.1 mm	0 to 4.9 mm	0 to 8.4 mm	0 to 1.4 mm	0 to 3.5 mm	0 to 7 mm
Differential travel		15% max. of sensing distance			10% max. of sensing distance		
Detectable object		Ferrous metals and non-ferrous metals (The sensing distance depends on the material of the sensing object. Refer to Engineering Data on catalog.)					
Standard sensing object (iron)		21 × 21 × 1 mm	30 × 30 × 1 mm	54 × 54 × 1 mm	12 × 12 × 1 mm	18 × 18 × 1 mm	30 × 30 × 1 mm
Response frequency *1		80 Hz	90 Hz	50 Hz	100 Hz	80 Hz	40 Hz
Power supply voltage		10 to 30 VDC (including 10% ripple (p-p)), Class 2					
Current consumption		1-output models (B1, B2, C1, C2): 16 mA max. 2-output models (B3, C3): 20 mA max.					
Output configuration		B□ Models: PNP open collector, C□ Models: NPN open collector					
Operation mode		1-output models (B1, C1): NO (Normally open), 1-output models (B2, C2): NC (Normally closed), 2-output models (B3, C3): NO+NC (Normally open, Normally closed)					
Control output	Load current	1-output models (B1, B2, C1, C2): 10 to 30 VDC, Class 2, 200 mA max. 2-output models (B3, C3): 10 to 30 VDC, Class 2, 100 mA max.					
	Residual voltage	1-output models (B1, B2, C1, C2): 2 V max. (Load current: 200 mA, Cable length: 2 m) 2-output models (B3, C3): 2 V max. (Load current: 100 mA, Cable length: 2 m)					
Indicator		Operation indicator (orange, lit) and communication indicator (green, not lit)					
Protection circuits		Power supply reverse polarity protection, Surge suppressor, Output short-circuit protection, Output reverse polarity protection					
Ambient temperature range		Operating: 0 to 85 °C, Storage: -15 to 85 °C (with no icing or condensation) *2					
Ambient humidity range		Operating/Storage: 35% to 95% (with no condensation)					
Temperature influence		±20% max. of sensing distance at 23 °C in the temperature range of 0 to 85 °C					
Voltage influence		±1.5% max. of sensing distance at rated voltage in the rated voltage ±15% range					
Insulation resistance		50 MΩ min. (at 500 VDC) between current-carrying parts and case					
Dielectric strength		1,000 VAC, 50/60 Hz for 1 minute between current-carrying parts and case					
Vibration resistance (destruction)		10 to 55 Hz, 1.5-mm double amplitude for 2 hours each in X, Y, and Z directions					
Shock resistance (destruction)		1,000 m/s ² 10 times each in X, Y, and Z directions					
Degree of protection		IEC 60529: IP67					
Connection method		Pre-wired Models (Standard cable length: 2 m), Pre-wired Connector Models (Standard cable length: 0.3 m)					
Weight (packed state)	Pre-wired	Approx. 140 g	Approx. 165 g	Approx. 225 g	Approx. 140 g	Approx. 160 g	Approx. 225 g
	M12 Pre-wired Smartclick Connector	Approx. 70 g	Approx. 100 g	Approx. 160 g	Approx. 70 g	Approx. 95 g	Approx. 160 g
Materials	Case	E2EW-X□: Stainless steel (SUS303), E2EW-QX□: Fluororesin coating (Base material: (SUS303))					
	Sensing surface	E2EW-X□: Stainless steel (SUS303), E2EW-QX□: Fluororesin coating (Base material: (SUS303))					
	Sensing surface (Thickness)	0.4 mm	0.4 mm	0.5 mm	0.8 mm	0.8 mm	0.8 mm
	Clamping nuts	E2EW-X□: Stainless steel (SUS303), E2EW-QX□: Fluororesin coating (Base material: (SUS303))					
	Toothed washers	Zinc-plated iron					
	Cable	Vinyl chloride (PVC)					
Accessories		Instruction manual, Clamping nuts, Toothed washer					

8. Proximity Sensor



Size		M8	M12	M18
Shielded		Shielded		
Model		E2E-X1R50□	E2E-X2R50□	E2E-X50□
Sensing distance		1.5 mm ±10%	2.5 mm ±10%	5 mm ±10%
Setting distance *1		0 to 1.2 mm	0 to 2 mm	0 to 4 mm
Differential travel		10% max. of sensing distance		
Detectable object		Ferrous metal (The sensing distance decreases with non-ferrous metal. Refer to Engineering Data on Catalog.)		
Standard sensing object		Iron, 10 × 10 × 1 mm	Iron, 12 × 12 × 1 mm	Iron, 18 × 18 × 1 mm
Response frequency *2		250 Hz	250 Hz	250 Hz
Power supply voltage		10 to 30 VDC, (including 10% ripple (p-p))		
Leakage current		0.8 mA max.		
Control output	Load current	3 to 100 mA		
	Residual voltage	Polarity: 3 V max. (Load current: 100 mA, Cable length: 2 m) No polarity: 5 V max. (Load current: 100 mA, Cable length: 2 m)		
Indicator		D1 Models: Operation indicator (orange), Setting indicator (green) D2 Models: Operation indicator (orange)		
Operation mode		D1 Models: NO D2 Models: NC Refer to the timing charts under I/O Circuit Diagrams on Catalog for details.		
Protection circuits		Surge suppressor, Load short-circuit protection		
Ambient temperature range		Operating: -25 to 70°C, Storage: -40 to 85°C (with no icing or condensation)		
Ambient humidity range		Operating and Storage: 35% to 95% (with no condensation)		
Temperature influence		±10% max. of sensing distance at 23°C in the temperature range of -25 to 70°C		
Voltage influence		±1% max. of sensing distance at rated voltage in the rated voltage ±15% range		
Insulation resistance		50 MΩ min. (at 500 VDC) between current-carrying parts and case		
Dielectric strength		1,000 VAC, 50/60 Hz for 1 minute between current-carrying parts and case		
Vibration resistance (destruction)		10 to 55 Hz, 1.5 mm double amplitude for 2 hours each in X, Y, and Z directions		
Shock resistance (destruction)		500 m/s ² 10 times each in X, Y, and Z directions	1,000 m/s ² 10 times each in X, Y, and Z directions	
Degree of protection		Pre-wired Models/Pre-wired Connector Models: IP67 (IEC 60529), IP67G *3 (JIS C 0920 Annex 1) Passed OMRON's Oil-resistant Component Evaluation Standards *4 (Cutting oil type: specified in JIS K 2241:2000, Temperature: 35°C max.) and ISO 20653 (old standard: DIN 40050 PART9) IP68K		
Connecting method		Pre-wired Models (Standard cable length: 2 m) and Pre-wired Connector Models (Standard cable length: 0.3 m)		
Weight (packed state)	Pre-wired Models	Approx. 80 g	Approx. 70 g	Approx. 130 g
	Pre-wired Connector Models	Approx. 30 g	Approx. 40 g	Approx. 70 g
Materials	Case	Stainless steel (SUS303)	Nickel-plated brass	
	Sensing surface	Polybutylene terephthalate (PBT)		
	Clamping nuts	Nickel-plated brass		
	Toothed washer	Zinc-plated iron		
	Cable	Vinyl chloride (PVC)		
Accessories		Instruction manual, Clamping nuts, Toothed washer		

9. Liquid Level Sensor



Item			E2K-L13MC□	E2K-L26MC□
Applicable pipes	Materials		Non-metal	
	Size	Diameter	8 to 11 mm	12 to 26 mm
		Thickness	1 mm max.	1.5 mm max.
Detectable object			Liquid *	
Repeat accuracy			±0.2 mm max.	
Differential travel (Reference value, varies with pipe size and liquid.)			0.6 to 5 mm	0.3 to 3 mm
Power supply voltage (operating voltage range)			12 to 24 VDC (10.8 to 30 VDC), ripple (p-p): 10% max.	
Current consumption			12 mA max.	
Control output	Load current		100 mA max.	
	Residual voltage		1 V max. (Load current: 100 mA, Cable length: 2 m)	
Sensing liquid position			Indented mark position (For details, refer to Datasheet)	
Indicators			Detection indicator (orange)	
Ambient temperature range			Operating: 0 to 55°C (with no icing or condensation), Storage: -10 to 65°C (with no icing or condensation)	
Ambient humidity range			Operating/Storage: 25% to 85% (with no condensation)	
Temperature influence			±4 mm of detection level at 23°C in the temperature range of 0 to 55°C (with pure water or 20% saline solution) (±6 mm for E2K-L13MC□ with pure water and a pipe diameter of 8 mm)	
Voltage influence			±0.5 mm of detection level at the rated voltage in rated voltage ±10% range	
Insulation resistance			50 MΩ min. (at 500 VDC) between current-carrying parts and case	
Dielectric strength			500 VAC, 50/60 Hz for 1 min between current-carrying parts and case	
Vibration resistance			Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 hours each in X, Y, and Z directions	
Shock resistance			Destruction: 500 m/s ² 3 times each in X, Y, and Z directions	
Degree of protection			IP66 (IEC)	
Connection method			Pre-wired Models (Standard cable length: 2 m)	
Weight (packed state)			Approx. 70 g	
Materials	Case, Cover		Heat-resistant ABS	
	Cable clamp		NBR	
Accessories			Two bands, Four slip-proof tubes, One driver for sensitivity adjustment, Instruction manual	

10. Long Distance Capacitive Sensor



Model		E2K-C25M□1	E2K-C25M□2	E2K-C25MY1	E2K-C25MY2
Sensing distance *		25 mm			
Sensing distance adjustable range		3 to 25 mm			
Detectable object		Conductors and dielectrics			
Standard sensing object		Grounded metal plate: 50 × 50 × 1 mm			
Differential travel		15% max. of sensing distance (when adjusted to 25 mm ±10% with standard sensing object)			
Response frequency		70 Hz		10 Hz	
Power supply voltage (operating voltage range)		12 to 24 VDC (10 to 30 VDC), ripple (p-p): 10% max.		100 to 220 VAC (90 to 250 VAC), 50/60 Hz	
Current consumption		E and F Models: 10 mA max. at 12 VDC, 16 mA max. at 24 VDC			
Leakage current		Y Models: 1 mA max. at 100 VAC (50/60 Hz) with output turned OFF, 2 mA max. at 200 VAC (50/60 Hz) with output turned OFF			
Control output	Load current	200 mA max.		5 to 200 mA (resistive load)	
	Residual voltage	2 V max. (Load current: 200 mA, Cable length: 2 m)		Refer to Engineering Data on Data Sheet	
Indicators		Detection indicator (red)		Operation indicator (red)	
Operation mode (with sensing object approaching)		E1, F1, and Y1 Models: NO E2, F2, and Y2 Models: NC Refer to the timing charts under I/O Circuit Diagrams on Data Sheet for details.			
Protection circuits		Reverse polarity protection, Surge suppressor		Surge suppressor	
Ambient temperature range		Operating/Storage: -25 to 70°C (with no icing or condensation)			
Ambient humidity range		Operating/Storage: 35% to 95% (with no condensation)			
Temperature influence		±15% max. of sensing distance at 23°C in the temperature range of -10 to 55°C ±25% max. of sensing distance at 23°C in the temperature range of -25 to 70°C			
Voltage influence		±2% max. of sensing distance at the rated voltage in rated voltage ±15% range		±2% max. of sensing distance at the rated voltage in rated voltage +20%, -10% range at 100 VAC, ±20% range at 200 VAC	
Insulation resistance		50 MΩ min. (at 500 VDC) between current-carrying parts and case			
Dielectric strength		1,000 VAC, 50/60 Hz for 1 min between current-carrying parts and case		1,500 VAC, 50/60 Hz for 1 min between current-carrying parts and case	
Vibration		Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 hours each in X, Y, and Z directions			
Shock resistance		Destruction: 500 m/s ² 10 times each in X, Y, and Z directions			
Degree of protection		IEC 60529 IP66			
Connection method		Pre-wired Models (Standard cable length: 2 m)			
Weight (packed state)		Approx. 200 g			
Materials	Case	Heat-resistant ABS			
	Sensing surface				
Accessories		Mounting Bracket, M4 screws, Instruction manual			

11. Inductive Sensor



Model		E2EZ-X2D[]-N E2EZ-X2D[]-M1J E2EZ-X2D[]-M1GJ	E2EZ-X4D[]-N E2EZ-X4D[]-M1J E2EZ-X4D[]-M1GJ	E2EZ-X8D[]-N E2EZ-X8D[]-M1J E2EZ-X8D[]-M1GJ	E2EZ-X4C1 E2EZ-X4B1 E2EZ-X4B1-M1J	E2EZ-X8C1 E2EZ-X8B1 E2EZ-X8B1-M1J
Sensing distance		2 mm ±10%	4 mm ±10%	8 mm ±10%	4 mm ±10%	8 mm ±10%
Set distance *1		0 to 1.6 mm	0 to 3.2 mm	0 to 6.4 mm	0 to 3.2 mm	0 to 6.4 mm
Differential travel		20% max. of sensing distance				
Detectable object		Ferrous metal (The sensing distance decreases with non-ferrous metal. Refer to Engineering Data on Data Sheet.)				
Standard sensing object		Iron, 12 × 12 × 1 mm	Iron, 30 × 30 × 1 mm	Iron, 54 × 54 × 1 mm	Iron, 30 × 30 × 1 mm	Iron, 54 × 54 × 1 mm
Response frequency *2		200 Hz	100 Hz	30 Hz	12 Hz	8 Hz
Power supply voltage (operating voltage range)		12 to 24 VDC (10 to 30 VDC), ripple (p-p): 10% max.			12 to 24 VDC (10 to 30 VDC), ripple (p-p): 10% max.	
Current consumption		—			15 mA max.	
Leakage current		0.8 mA max.			—	
Control output	Load current	3 to 100 mA max.			NPN open-collector output/ PNP open-collector output 100 mA max. at 12 VDC (30 VDC max.) 200 mA max. at 24 VDC (30 VDC max.)	
	Residual voltage	3 V max. (Load current: 100 mA, Cable length: 2 m)			2 V max. (Load current: 200 mA, Cable length: 2 m)	
Indicators		D1 Models: Operation indicator (red), Setting indicator (green) D2 Models: Operation indicator (red)			Detection indicator (red)	
Operation mode (with sensing object approaching)		D1 Models: NO D2 Models: NC For details, refer to the Timing chart on Data Sheet.			NO For details, refer to the Timing chart on Data Sheet.	
Protection circuits		Load short-circuit protection, Surge suppressor			Load short-circuit protection, Reverse polarity protection, Surge suppressor	
Ambient temperature range		Operating/Storage: 0 to 50°C (with no icing or condensation)				
Ambient humidity range		Operating/Storage: 35% to 95% (with no condensation)				
Temperature influence		±20% max. of sensing distance at 23°C in the temperature range of 0 to 50°C				
Voltage influence		±2.5% max. of sensing distance at rated voltage in the rated voltage ±10% range				
Insulation resistance		50 MΩ min. (at 500 VDC) between current-carrying parts and case				
Dielectric strength		1,000 VAC, 50/60 Hz for 1 minute between current-carrying parts and case				
Vibration resistance		Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 hours each in X, Y, and Z directions				
Shock resistance		Destruction: 1,000 m/s ² 10 times each in X, Y, and Z directions				
Degree of protection		IEC 60529 IP67, in-house standards: oil-resistant				
Connection method		Pre-wired Models (Standard cable length: 2 m) and Pre-wired Connector Models				
Weight (packed state)		E2EZ-X2D[]-N: Approx. 70 g E2EZ-X2D[]-M1J: Approx. 40 g E2EZ-X2D[]-M1GJ: Approx. 40 g	E2EZ-X4D[]-N: Approx. 160 g E2EZ-X4D[]-M1J: Approx. 90 g E2EZ-X4D[]-M1GJ: Approx. 90 g	E2EZ-X8D[]-N: Approx. 220 g E2EZ-X8D[]-M1J: Approx. 160 g E2EZ-X8D[]-M1GJ: Approx. 160 g	Approx. 170 g	Approx. 270 g
Materials	Case	Nickel-plated brass				
	Sensing surface	PBT			Heat-resistant ABS	
	Clamping nuts	Zinc-plated iron				
	Toothed washer	Zinc-plated iron				
Accessories		Instruction manual				

12. Colour Sensor



Sensing method		Diffuse reflective (mark detection)
Model	NPN output	E3ZM-V61/-V66
	PNP output	E3ZM-V81/-V86
Sensing distance		12±2 mm *1
Sensing range		Depends on the combination of colors. Refer to Data Sheet.
Spot diameter		2-mm dia. max.
Light source (wavelength)		White LED (450 to 700 nm)
Power supply voltage		10 to 30 VDC, including 10% ripple (p-p)
Power consumption		750 mW max. (current consumption for a 30-V power supply voltage: 25 mA max.)
Control output		Load power supply voltage: 30 VDC max., Load current: 100 mA max. (Residual voltage: 2 V max.) Open-collector output (NPN/PNP output depending on model)
Remote control input		NPN output ON: Short-circuit to 0 V, or 1.5 V max. (source current: 1 mA max.) NPN output OFF: Open or Vcc - 1.5 V to Vcc (leakage current: 0.1 mA max.) PNP output ON: Vcc - 1.5 V to Vcc (sink current: 1 mA max.) PNP output OFF: Open or 1.5 V max. (leakage current: 0.1 mA max.)
Operating modes		Set in the order of the teaching operation. *2
Protection circuits		Reversed power supply polarity, Load short-circuit protection, and Reversed output polarity protection
Response time		Operate or reset: 50 µs max.
Sensitivity adjustment		Teaching method
Ambient illumination		(Receiver side) Incandescent lamp: 3,000 lx max., Sunlight: 10,000 lx max.
Ambient temperature range		Operating: -40 to 60°C (*3), Storage: -40 to 70°C (with no icing or condensation)
Ambient humidity range		Operating: 35% to 85%, Storage: 35% to 95% (with no condensation)
Insulation resistance		20 MΩ min. (at 500 VDC)
Dielectric strength		1,000 VAC at 50/60 Hz for 1 min
Vibration resistance (destruction)		10 to 55 Hz, 1.5-mm double amplitude for 2 h each in X, Y, and Z directions
Shock resistance (destruction)		500 m/s ² for 3 times each in X, Y, and Z directions
Degree of protection *4		IEC 60529: IP67, DIN 40050-9: IP69K
Connection method		Pre-wired cable (standard length: 2 m) or M8 4-pin connector
Indicator		Operating indicator (yellow), Stability indicator (green), and Teaching indicator (red)
Weight (packed state)		Pre-wired models (2-m cable): Approx. 85 g Connector models: Approx. 35 g
Materials	Housing	SUS316L
	Lens	PMMA (polymethylmethacrylate)
	Indication	PES (polyethersulfone)
	Buttons	Fluoro rubber
	Cable	PVC (polyvinyl chloride)
Accessories		Instruction sheet Note: Mounting Brackets are purchased separately.

13. Chemical Resistance Capacitive Sensor



Model		E2KQ-X10ME1	E2KQ-X10ME2
Sensing distance *1		10 mm	
Sensing distance adjustable range		6 to 10 mm	
Differential travel		4% to 20% of sensing distance	
Detectable object		Conductors and dielectrics	
Standard sensing object		Grounded metal plate: 50 × 50 × 1 mm	
Response frequency		35 Hz	
Power supply voltage (operating voltage range)		12 to 24 VDC (10 to 30 VDC), ripple (p-p): 10% max.	
Current consumption		15 mA max.	
Control output	Load current	100 mA	
	Residual voltage	1.5 V max. (Load current: 100 mA, Cable length: 2 m)	
Indicators		Detection indicator (red)	
Operation mode (with sensing object approaching) *2		NO	NC
Protection circuits		Reverse polarity protection, Surge suppressor	
Ambient temperature range		Operating: -10 to 55°C, Storage: -25 to 55°C (with no icing or condensation)	
Ambient humidity range		Operating/storage: 35% to 85% (with no condensation)	
Temperature influence		±15% max. of sensing distance at 23°C in the temperature range of -10 to 55°C	
Voltage influence		±2% max. of sensing distance at rated voltage at rated voltage ±20%	
Insulation resistance		50 MΩ min. (at 500 VDC) between current-carrying parts and case	
Dielectric strength		500 VAC, 50/60 Hz for 1 min between current-carrying parts and case	
Vibration resistance		Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 hours each in X, Y, and Z directions	
Shock resistance		Destruction: 500 m/s ² 3 times each in X, Y, and Z directions	
Degree of protection		IP66 (IEC), in-house standards: oil-resistant	
Connection method		Pre-wired Models (Standard cable length: 2 m)	
Weight (packed state)		Approx. 150 g	
Materials	Case, sensing surface	Fluorine resin	
	Clamping nuts		
	Cable	Vinyl chloride	
Accessories		Adjustment screwdriver, Instruction manual	

14. Flat Type Capacitive Sensor



Model		E2K-F10MC□-A	E2K-F10MC□
Sensing distance		10 mm (Sensing distance adjustable range: 4 to 10 mm)	10 mm ±10%
Set distance		0 to 7.5 mm *	
Differential travel		15% max. of sensing distance	
Detectable object		Conductors and dielectrics	
Standard sensing object		Grounded metal plate: 50 × 50 × 1 mm	
Response frequency		100 Hz	
Power supply voltage (operating voltage range)		12 to 24 VDC (10 to 30 VDC), ripple (p-p): 10% max.	
Current consumption		10 mA max. at 24 VDC	
Control output	Load current	NPN open collector, 100 mA max. (at 30 VDC)	
	Residual voltage	1.5 V max. (Load current: 100 mA, Cable length: 2 m)	
Indicators		Detection indicator (red)	
Number of turns of sensitivity adjustment		11 turns	-
Operation mode (with sensing object approaching)		NO (Refer to the timing charts under I/O Circuit Diagrams on Data Sheet for details.)	
Protection circuits		Reverse polarity protection, Surge suppressor	
Ambient temperature range		Operating/Storage: -10 to 55°C (with no icing or condensation)	
Ambient humidity range		Operating/Storage: 35% to 95%	Operating/Storage: 35% to 95%
Temperature influence		±15% max. of sensing distance at 23°C in the temperature range of -10 to 55°C	
Voltage influence		±2.5% max. of sensing distance at rated voltage at rated voltage ±10%	
Insulation resistance		50 MΩ min. (at 500 VDC) between current-carrying parts and case	
Dielectric strength		500 VAC, 50/60 Hz for 1 min between current-carrying parts and case	
Vibration resistance		Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 hours each in X, Y, and Z directions	
Shock resistance		Destruction: 500 m/s ² 3 times each in X, Y, and Z directions	
Degree of protection		IP64 (IEC)	IP66 (IEC)
Connection method		Pre-wired Models (Standard cable length: 2 m)	
Weight (packed state)		Approx. 35 g	
Materials	Case	Heat-resistant ABS	
	Sensing surface		
Accessories		Adjustment screwdriver, Instruction manual	

15. Long Distance Capacitive Sensor



Model		E2J-W10MA	E2J-W20MA
Sensing distance		10 mm	20 mm
Sensing distance adjustable range		4 to 10 mm	8 to 20 mm
Differential travel		15% max. of sensing distance	
Detectable object		Conductors and dielectrics	
Standard sensing object		Grounded metal plate: 50 × 50 × 1 mm	
Response frequency		70 Hz min.	
Indicators		Detection indicator (yellow)	
Ambient temperature range		Operating/Storage: -10 to 55°C (with no icing or condensation)	
Ambient humidity range		Operating/Storage: 35% to 85% (with no condensation)	
Vibration resistance		Destruction: 10 to 500 Hz, 2-mm double amplitude or 150 m/s ² for 2 hours each in X, Y, and Z directions	
Shock resistance		Destruction: 500 m/s ² 3 times each in X, Y, and Z directions	
Degree of protection		IP66 (IEC)	
Connection method		Pre-wired Connector Models (Robotics cable, Standard cable length: 1m)	
Weight (packed state)		Approx. 30 g	Approx. 40 g
Materials	Case	Heat-resistant ABS	

16. Measurement Sensor



Model	ZX-GT28S11	ZX-GT2840S11	ZX-GT28S41	ZX-GT2840S41
Output type	NPN		PNP	
Appearance	Separate type	Integrated type	Separate type	Integrated type
Light source	Visible semiconductor laser diode (wavelength 650 nm, CLASS 1 of EN60825-1/ IEC60825-1, CLASS II of FDA (21CFR 1040.10 and 1040.11))			
Measuring width	28 mm			
Sensing distance	0 to 500 mm	40 mm	0 to 500 mm	40 mm
Minimum sensing object	0.5 mm dia. *1	0.2 mm dia.	0.5 mm dia. *1	0.2 mm dia.
Linearity	±0.1%F.S. *2			
Resolution	10 μm (number of process values to average: 16) *3			
Temperature characteristic	0.01%F.S./°C *4			
Indicators (emitter)	Laser ON indicator (green), laser alarm indicator (red)			
Indicators (receiver)	Optical axis setting indicator (green)			
Laser OFF input/sync input	ON: Short-circuited with 0 V or 1.5 V max. OFF: Open (leakage current: 0.1 mA max.)		ON: Short-circuited with power supply voltage or power supply voltage -1.5 V max. OFF: Open (leakage current: 0.1 mA max.)	
Laser deterioration alarm output	NPN open-collector output 30 VDC 20 mA max. Residual voltage 1.2 V max.		PNP open-collector output 30 VDC 20 mA max. Residual voltage 2 V max.	
Power consumption (emitter)	30 mA max.			
Power supply voltage (emitter)	24 VDC +10%, -15% ripple (p-p) 10% max.			
Dielectric strength	1,000 VAC, 50/60 Hz for 1 min			
Insulation resistance	20 MΩ (at 500 VDC)			
Operating ambient illumination (emitter)	3,000 lx (incandescent light)			
Operating ambient illumination (receiver)	1,000 lx (incandescent light) *5			
Ambient temperature	Operating: 0 to 40°C Storage: -15 to 50°C (with no icing or condensation)			
Ambient humidity	Operating and storage: 35% to 85% (with no condensation)			
Vibration resistance (durability)	10 to 150 Hz Single-amplitude: 0.75 mm for 80 min each in X, Y and Z directions			
Shock resistance (durability)	300 m/s ² 3 times each in six directions (up/down, left/right, forward/backward)			
Degree of protection	IEC60529 IP40			
Cable length	2 m			
Material	Case: aluminum die-cast, Lens: glass			
Weight (packed state)	Approx. 550 g	Approx. 570 g	Approx. 550 g	Approx. 570 g
Accessories	Laser warning labels, Instruction Sheet			

17. Photomicro Sensor



Sensing method	Through-beam type (with slot)	Reflective type
Models	EE-SPX301, EE-SPX401	EE-SPY301, EE-SPY401 EE-SPY302, EE-SPY402
Sensing distance	3.6 mm (slot width)	5 mm (Reflection factor: 90%; white paper 15 × 15 mm) *1
Sensing object	Opaque: 1 × 0.5 mm min.	---
Differential distance	0.05 mm max.	0.2 mm max. (with a sensing distance of 3 mm, horizontally)
Light source	GaAs infrared LED with a peak wavelength of 940 nm	
Indicator *2	Light indicator (red)	
Supply voltage	5 to 24 VDC ±10%, ripple (p-p): 5% max.	
Current consumption	Average: 15 mA max., Peak: 50 mA max.	
Control output	NPN voltage output: Load power supply voltage: 5 to 24 VDC Load current: 80 mA max. OFF current: 0.5 mA max. 80 mA load current with a residual voltage of 1.0 V max. 10 mA load current with a residual voltage of 0.4 V max.	
Response frequency *3	500 Hz min.	100 Hz min.
Ambient illumination	3,000 lx max. with incandescent light or sunlight on the surface of the receiver	
Ambient temperature range	Operating: - 10 to +55 ° C Storage: - 25 to +65 ° C (with no icing)	
Ambient humidity range	Operating: 5% to 85% Storage: 5% to 95% (with no condensation)	
Vibration resistance	Destruction: 10 to 55 Hz, 1.5-mm double amplitude for 2 h each in X, Y, and Z directions	
Shock resistance	Destruction: 500 m/s ² for 3 times each in X, Y, and Z directions	
Degree of protection	IEC IP50	
Connecting method	Special connector (soldering not possible)	
Weight	Approx. 2.6 g	
Material	Case	Polycarbonate

18. Pipe Mounting Liquid Detector Sensor



Models		EE-SPX613
Applicable pipe	Any 6- to 13-mm-diameter pipe with a wall thickness of 1 mm that is made of FEP or any other material as transparent as FEP.	
Sensing object	Liquids in pipes (High-viscosity liquids or liquids with floating materials may not be detected.)	
Light source	GaAs infrared LED with a peak wavelength of 940 nm	
Indicator	Light indicator GaP (Red LED: Peak wavelength of 700 nm)	
Supply voltage	12 to 24 VDC $\pm 10\%$, ripple (p-p): 5% max.	
Current consumption	Average: 30 mA max., Peak: 80 mA max.	
Control output	NPN open collector: Load power supply voltage: 5 to 24 VDC Load current: 100 mA max. OFF current: 0.5 mA max. 100 mA load current with a residual voltage of 0.8 V max. 40 mA load current with a residual voltage of 0.4 V max.	
Ambient illumination	3,000 lx max. with incandescent light or sunlight on the surface of the receiver	
Ambient temperature range	Operating: - 10 to +55 °C Storage: - 25 to +65 °C (with no icing or condensation)	
Ambient humidity range	Operating: 5% to 85% Storage: 5% to 95% (with no condensation)	
Vibration resistance	Destruction: 10 to 500 Hz, 1.0-mm single amplitude or 150 m/s ² in X, Y, and Z directions 3 times and for 11 min each	
Shock resistance	Destruction: 500 m/s ² for 3 times each in X, Y, and Z directions	
Degree of protection	IEC 60529 IP50	
Connecting method	Pre-wired (Standard length: 1 m)	
Weight (packed state)	Approx. 55 g	
Material	Case	Polycarbonate
	Cover	
Accessories	Support belt (2), slip protection tube (2), Instruction Manual	

19. Fiber Sensor



Model	Installation			Cable						Weight (packed state)
	Ambient temperature	Tightening torque	Mounting hole	Bending radius	Unbendable length	Tensile strength	Sheath material	Core material	Emitter/receiver differentiation	
E32-LT35Z 2M	-40 to 70°C	0.15 N·m	-	R1	0	9.8 N	Polyethylene	Plastic	None	Approx. 25 g

20. Area Picking Sensor

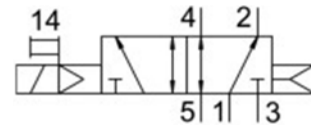


Sensing method Model		Through-beam	
		F3W-E032A6/A8	F3W-E032B6/B8
Sensing distance		30 to 300 mm	
Beam gap		25 mm	
Number of beams		3	
Sensing width		50 mm	
Standard sensing object		Opaque, 28-mm dia. min.	
Light source (emission wavelength)		Infrared LED (860 nm)	
Power supply voltage		12 to 24 VDC $\pm 10\%$ (ripple (p-p): 10% max.)	
Current consumption		Emitter: 40 mA max., Receiver: 40 mA max.	
Control output		NPN or PNP open collector output Load power supply: 30 VDC, Load current: 100 mA max. (Residual voltage: 1.8 V max.) Dark-ON or Light-ON (selectable)	
Picking instruction indicator input		Relay or open collector input NPN input Indicator ON: Input voltage of 0 to 2 V Indicator OFF: Open (with leakage current of 0.1 mA max.) PNP input Indicator ON: Input voltage: (+DC - 2 V) to +DC * Indicator OFF: Open (with leakage current of 0.1 mA max.)	
Emission stop input		Relay or open collector input NPN input Emission stop input voltage: 0 to 2 V Emission input voltage: Open (with leakage current of 0.1 mA max.) PNP input Emission stop input voltage: (+DC - 2 V) to +DC * Emission input voltage: Open (with leakage current of 0.1 mA max.)	
Protection circuits		Reverse-connection protection, output short protection, and mutual interference prevention function (set with frequency switch)	
Response time		Operate/Reset: 100 ms max.	
Indicators	Receiver	Operation indicator (orange), stability indicator (green), and picking indicators (orange)	
	Emitter	Power indicator (green), emission stop indicator (orange), and picking indicators (orange)	
Ambient temperature		Operating: -10 to 55 °C, Storage: -25 to 70 °C (with no icing or condensation)	
Ambient humidity		Operating/storage: 35% to 85% (with no condensation)	
Ambient illumination		Sunlight: 10,000 lx at light-receiving surface, Incandescent light: 3,000 lx at light-receiving surface	
Insulation resistance		20 M Ω min. (at 500 VDC)	
Dielectric strength		1,000 VAC, 50/60 Hz for 1 min	
Vibration resistance (destruction)		10 to 50 Hz, 1.5-mm double-amplitude for 2 hours each in X, Y, and Z directions	
Shock resistance (destruction)		500 m/s ² , 3 times in X, Y, and Z directions	
Degree of protection		IP62 (IEC60529)	
Connection method		Pre-wired Standard cable length: 2 m	Pre-wired connector Standard cable length: 0.3 m (M12, 4-pin connector)
		Approx. 125 g	Approx. 85 g
Weight (packed state)		Approx. 125 g	Approx. 85 g
Materials	Case	ABS resin	
	Lens	Acrylic resin	
	Cable	Oil-resistant PVC	
Accessories		Instruction manual	

I. Jenis actuator

1. Solenoid Valve

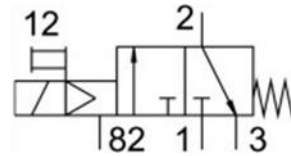
Festo VUVG-L14-M52-AT-G18-1P3



Feature	Value
Valve function	5/2-way, monostable
Type of actuation	Electric
Valve size	14 mm
Standard nominal flow rate	700 l/min ... 780 l/min
pneumatic working port	G1/8
Operating voltage	24V DC
Operating pressure	0.25 MPa ... 0.8 MPa
Operating pressure	2.5 bar ... 8 bar
Design	Piston gate valve
Type of reset	Pneumatic spring
Approval	RCM trademark c UL us - Recognized (OL)
Degree of protection	IP40 IP65 With plug socket
Nominal size	5.6 mm
Exhaust-air function	With flow control option
Sealing principle	Soft
Mounting position	optional
Manual override	Detenting Non-detenting Covered
Type of piloting	Pilot actuated
Pilot air supply	Internal
Symbol	00992908
lap	Positive overlap
Pilot pressure	0.25 MPa ... 0.8 MPa
Pilot pressure	2.5 bar ... 8 bar
Suitability for vacuum	no
Switching time off	22 ms
Switching time on	14 ms
Duty cycle	100%
Max. positive test pulse with 0 signal	700 µs
Max. negative test pulse with 1 signal	900 µs
Characteristic coil data	24 V DC: 1.0 W 24 V DC: low-current phase 0.3 W, high-current phase 1.0 W
Permissible voltage fluctuations	+/- 10 %

Operating medium	Compressed air to ISO 8573-1:2010 [7:4:4]
Note on operating and pilot medium	Lubricated operation possible (in which case lubricated operation will always be required)
Vibration resistance	Transport application test with severity level 2 to FN 942017-4 and EN 60068-2-6
Restrictions for environmental and media temperature	-5 ... 50° C Without holding current reduction
Shock resistance	Shock test with severity level 2 to FN 942017-5 and EN 60068-2-27
Corrosion resistance class CRC	2 - Moderate corrosion stress
LABS (PWIS) conformity	VDMA24364-B1/B2-L
Media temperature	-5 °C ... 60 °C
Ambient temperature	-5 °C ... 60 °C
Product weight	78 g
Electrical connection	Via electrical sub-base
Type of mounting	Either: On manifold rail With through-hole
Pneumatic connection, port 1	G1/8
Pneumatic connection, port 2	G1/8
Pneumatic connection, port 3	G1/8
Pneumatic connection, port 4	G1/8
Pneumatic connection, port 5	G1/8
Note on materials	RoHS-compliant
Material seals	HNBR NBR
Material housing	Wrought aluminium alloy

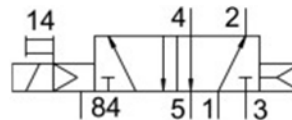
2. Festo MFH-3-1/4



Feature	Value
Valve function	3/2-way, closed, monostable
Type of actuation	Electric
Construction width	30.5 mm
Standard nominal flow rate	800 l/min
pneumatic working port	G1/4
Operating voltage	Via solenoid coil, must be ordered separately
Operating pressure	0.15 MPa ... 0.8 MPa
Operating pressure	1.5 bar ... 8 bar
Design	Poppet seat
Type of reset	Mechanical spring
Approval	c UL us - Recognized (OL)
Degree of protection	IP65
Nominal size	7 mm
Grid dimension	32 mm
Exhaust-air function	With flow control option
Sealing principle	Soft
Mounting position	optional

Manual override	Detenting
Type of piloting	Pilot actuated
Flow direction	Non-reversible
Symbol	00991041
lap	Underlap
Switching time off	29 ms
Switching time on	10 ms
Max. positive test pulse with 0 signal	2200 µs
Max. negative test pulse with 1 signal	3700 µs
Characteristic coil data	See solenoid coil, to be ordered separately
Operating medium	Compressed air to ISO 8573-1:2010 [7:4:4]
Note on operating and pilot medium	Lubricated operation possible (in which case lubricated operation will always be required)
Corrosion resistance class CRC	1 - Low corrosion stress
LABS (PWIS) conformity	VDMA24364-B1/B2-L
Storage temperature	-20 °C ... 60 °C
Media temperature	-10 °C ... 60 °C
Ambient temperature	-5 °C ... 40 °C
Product weight	320 g
Electrical connection	Via F coil, to be ordered separately
Type of mounting	Either: On manifold rail With through-hole
Pilot exhaust port 82	M5
Pneumatic connection, port 1	G1/4
Pneumatic connection, port 2	G1/4
Pneumatic connection, port 3	G1/4
Note on materials	RoHS-compliant
Material seals	NBR
Material housing	Die-cast aluminium

3. Festo CPE18-M1H-5L-1/4



Feature	Value
Valve function	5/2-way, monostable
Type of actuation	Electric
Construction width	18 mm
Standard nominal flow rate	1500 l/min
pneumatic working port	G1/4
Operating voltage	24V DC
Operating pressure	0.25 MPa ... 1 MPa
Operating pressure	2.5 bar ... 10 bar
Design	Piston gate valve
Type of reset	Pneumatic spring
Approval	c UL us - Recognized (OL)
Maritime classification	See certificate
Degree of protection	IP65 With plug socket To IEC 60529

Nominal size	8 mm
Exhaust-air function	With flow control option
Sealing principle	Soft
Mounting position	optional
Manual override	Detenting via accessory Non-detenting
Type of piloting	Pilot actuated
Pilot air supply	Internal
Flow direction	Non-reversible
Symbol	00991023
Valve position code	Inscription label holder
lap	Positive overlap
Switching time off	20 ms
Switching time on	26 ms
Duty cycle	100%
Max. positive test pulse with 0 signal	3300 µs
Max. negative test pulse with 1 signal	3100 µs
Characteristic coil data	24 V DC: 1.5 W
Permissible voltage fluctuations	-15%/+10%
Operating medium	Compressed air to ISO 8573-1:2010 [7:4:4]
Note on operating and pilot medium	Lubricated operation possible (in which case lubricated operation will always be required)
Vibration resistance	Transport application test with severity level 2 to FN 942017-4 and EN 60068-2-6
Shock resistance	Shock test with severity level 2 to FN 942017-5 and EN 60068-2-27
Corrosion resistance class CRC	2 - Moderate corrosion stress
LABS (PWIS) conformity	VDMA24364-B1/B2-L
Media temperature	-5 °C ... 50 °C
Ambient temperature	-5 °C ... 50 °C
Product weight	220 g
Electrical connection	Type C
Type of mounting	With through-hole
Pilot exhaust port 82	M5
Pilot exhaust port 84	M5
Pilot air port 12	M5
Pilot air port 14	M5
Pneumatic connection, port 1	G1/4
Pneumatic connection, port 2	G1/4
Pneumatic connection, port 3	G1/4
Pneumatic connection, port 4	G1/4
Pneumatic connection, port 5	G1/4
Note on materials	RoHS-compliant
Material seals	NBR
Material housing	Die-cast aluminium

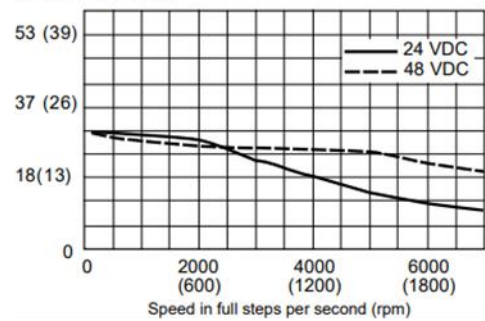
4. Stepper Motor NEMA 17 M-1713-1.5



1.5 Amp motors		Single length
Part number		M-1713-1.5 • (1)
Holding torque	oz-in	32
	N-cm	23
Detent torque	oz-in	1.7
	N-cm	1.2
Rotor inertia	oz-in-sec ²	0.000538
	kg-cm ²	0.038
Weight	oz	7.4
	grams	210
Phase current	amps	1.5
Phase resistance	ohms	1.3
Phase inductance	mH	2.1

M-1713-1.5

Torque in oz-in (N-cm)



5. Autronics A3K-S545



Sudut step dasar [FULL/HALF]	0.72°/0.36°
Maks. Torsi Holding	2.4kgf.cm
Momen Rotor Inersia	68g·cm ²
Ketahanan Lilitan	2.2Q
Kisaran arus	0.75A/Phase
Panjang Motor	47mm
Tipe Shaft	Fasa tunggal
Koneksi kabel	Pentagon

Kisaran arus	0.75A/Phase
Panjang Motor	47mm
Tipe Shaft	Fasa tunggal
Koneksi kabel	Pentagon
Tingkat proteksi	IP30
Rem Elektromagnetik_Tegangan Pemicu	24VDC $\pm 10\%$ (no-polarity)
Rem Elektromagnetik_Arus Pemicu	0.2A
ElectroMagnetic brake_Torsi Static friction	1.8kgf.cm
ElectroMagnetic brake_rotasi momen inersia	3×10^{-7} kgf.cm ²
Rem Elektromagnetik_Tipe Rem B	Power on: brake is released, power off: brake is operating
Rem Elektromagnetik_Waktu Pengoperasian	Max. 25ms
Rem Elektromagnetik_Waktu Pelepasan	Max. 15ms
Berat	Kurang lebih 0.49kg (kurang lebih 0.4kg)

6. Schneider BRS366H030AAA



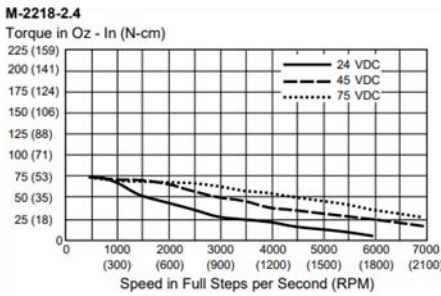
Range compatibility	Lexium SD3
Product or component type	Motion control motor
Device short name	BRS3
Maximum mechanical speed	3000 rpm
Motor type	3-phase stepper motor
Number of motor poles	6
Supply voltage limits	34 V DC 48 V DC
Mounting support	Flange
Motor flange size	57.2 mm
Length	56 mm
Centring collar diameter	38 mm

Maximum starting frequency	8 kHz
[In] rated current	5.8 A
Resistance	0.5 Ohm (winding)
Time constant	3.3 ms
Maximum radial force Fr	40 N (second shaft end) 24 N (first shaft end)
Maximum axial force Fa	100 N (tensile force) 8.4 N (force pressure)
Service life in hours	20000 h (bearing)
Angular acceleration	200000 rad/s ²
Net weight	1.6 kg
Shaft diameter	6.35 mm
Shaft length	21 mm
Nominal torque	0.9 N.m
Holding torque	1.02 N.m
Rotor inertia	0.22 kg.cm ²
Resolution	1.8 °, 0.9 °, 0.72 °, 0.36 °, 0.18 °, 0.09 °, 0.072 °, 0.036 ° step angle 200, 400, 500, 1000, 2000, 4000, 5000, 10000 steps number of full steps per revolution
Accuracy error	+/- 6 arc min
Centring collar depth	1.6 mm
Number of mounting holes	4
Mounting holes diameter	5.2 mm
Circle diameter of the mounting holes	66.6 mm
Electrical connection	Flying leads
Holding brake	Without
Shaft end	Smooth shaft
Second shaft	Without second shaft end

7. Schneider NEMA 23 M-2218-2.4S



2.4 Amp motors		Single length
Part number		M-2218-2.4S (1)
Holding torque	oz-in	90
	N-cm	64
Detent torque	oz-in	3.9
	N-cm	2.7
Rotor inertia	oz-in-sec ²	0.00255
	kg-cm ²	0.18
Weight	oz	16.9
	grams	480
Phase current	amps	2.4
Phase resistance	ohms	0.95
Phase inductance	mH	2.4



8. FESTO EMMS-ST-57-S-S-G2



Feature	Value
Ambient temperature	-10 °C ... 50 °C
Storage temperature	-20 °C ... 70 °C
Relative air humidity	0 - 85%
Conforms to standard	IEC 60034
Insulation protection class	B
Temperature class as per EN 60034-1	B
Rating class as per EN 60034-1	S1
Degree of protection	IP54
Electrical connector system	Plug
Note on materials	RoHS-compliant
Corrosion resistance class CRC	1 - Low corrosion stress
LABS (PWIS) conformity	VDMA24364-B2-L
Approval	RCM trademark c UL us - Recognized (OL)
CE mark (see declaration of conformity)	To EU EMC Directive In accordance with EU RoHS Directive
CE marking (see declaration of conformity)	To UK instructions for EMC To UK RoHS instructions
Nominal operating voltage DC	48 V
Motor holding torque	0.8 Nm
Max. rotational speed	2720 rpm
Stepper angle for complete step	1.8 deg
Stepping angle tolerance	±5%
Nominal motor current	5 A
Voltage constant, phase	12 mV/min
Phase winding resistance	0.15 Ohm
Phase winding inductance	0.5 mH
Mass moment of inertia of rotor	0.29 kgcm ²
Total mass moment of inertia of output	0.29 kgcm ²
Product weight	870 g
Permissible axial shaft load	10 N
Permissible radial shaft load	52 N
Brake holding torque	0.4 Nm

9. Servo Motor

FESTO EMME-AS-40-M-LV-AMB



Feature	Value
Nominal operating voltage DC	360 V
Nominal voltage DC	360 V
Type of winding switch	Star inside
Number of pole pairs	2
Standstill torque	0.35 Nm
Nominal torque	0.21 Nm
Peak torque	1.4 Nm
Nominal rotary speed	9000 rpm
Max. rotational speed	10000 rpm
Nominal power rating of motor	200 W
Continuous stall current	1.6 A
Nominal motor current	1.2 A
Peak current	6.4 A
Motor constant	0.175 Nm/A
Voltage constant, phase-to-phase	13.2 mV/min
Phase-phase winding resistance	8.6 Ohm
Phase-phase winding inductance	4.51 mH
Total mass moment of inertia of output	0.079 kgcm ²
Product weight	850 g
Permissible axial shaft load	12 N
Permissible radial shaft load	115 N
Rotor position sensor	Absolute multi-turn encoder
Rotor position encoder interface	HIPERFACE®
Rotor position sensor, encoder measuring principle	Capacitive
Rotor pos. enc., sin/cosin p/r	16
Rotor pos. encoder, typ. res.	12 bit
Rot. pos. enc., typ. ang. acc.	20 arcmin
Brake holding torque	0.4 Nm
Operating voltage DC for brake	24 V
Power consumption, brake	8 W
Mass moment of inertia of brake	0.014 kgcm ²
Switching cycles holding brake	5 million idle actuations (without friction work!)
Mean time to failure (MTTF), subcomponent	371 years, holding brake
Mean time to dangerous failure (MTTFd), subcomponent	271 years, rotor position sensor

10. Autonics Ai-M-42MA-G7.2



Tipe motor	Tipe gear bawaan
Rasio Gear	1:7.2
Maks. Torsi Holding	2N.m
Momen Rotor Inersia	54×10^{-7} kg.m ²
Kisaran arus	1.7A/Phase
Torsi yang digunakan	1.5N.m
Sudut step standar	0.25 °
Backlash	35min (0.58 °)
Ketahanan Lilitan	1.85Q
Induktansi	3.5 mH/Fasa
Berat	≈ 0.58 kg (≈ 0.70kg)
Fasa Motor	2 fasa
Motor_Metode Operasi	Bipolar
Kelas insulasi	Tipe B (130 °C)
Ketahanan insulasi	≥ 100 MΩ (di antara motor coil-case 500 VDC megger)
Dielectric strength	500 VAC ~ 50/60 Hz selama 1 menit (di antara motor coil-case)

Vibrasi	Amplitudo 1.5mm pada frekuensi dari 10 ~ 55Hz (selama 1 menit) di setiap arah X, Y, Z selama 2 jam
Getaran	≤ 50 G
Lingkungan_kadar suhu	0 hingga 50°C, ruang tertutup: -20 hingga 70°C
Lingkungan_kadar kelembaban	20 ~ 85%RH, ruang tertutup: 15 ~ 90%RH
Tingkat proteksi	IP30(standar IEC)
Catu daya	5VDC ±5%(ripple P-P: maks. 5%)
Konsumsi arus	≤ 50 mA (tanpa beban)
Resolusi	10,000PPR(2,500PPR×4-multiply)
Output kontrol	Line Driver
Fasa output	A, $\overline{A}$, B, $\overline{B}$, Z, $\overline{Z}$
Maks. Frekuensi respon	300kHz

11. Kontaktor Magnetik Schneider LC1D



Range of product	TeSys Deca
Product or component type	Contacteur
Device short name	LC1D
Contacteur application	Resistive load Motor control
Utilisation category	AC-4 AC-1 AC-3 AC-3e
Poles description	3P
[Ue] rated operational voltage	Power circuit: <= 690 V AC 25...400 Hz Power circuit: <= 300 V DC
[Ie] rated operational current	32 A (at <60 °C) at <= 440 V AC AC-3 for power circuit 50 A (at <60 °C) at <= 440 V AC AC-1 for power circuit 32 A (at <60 °C) at <= 440 V AC AC-3e for power circuit
[Uc] control circuit voltage	220 V AC 50/60 Hz

Complementary

Motor power kW	7.5 kW at 220...230 V AC 50/60 Hz (AC-3) 15 kW at 380...400 V AC 50/60 Hz (AC-3) 15 kW at 415...440 V AC 50/60 Hz (AC-3) 18.5 kW at 500 V AC 50/60 Hz (AC-3) 18.5 kW at 660...690 V AC 50/60 Hz (AC-3) 7.5 kW at 400 V AC 50/60 Hz (AC-4) 7.5 kW at 220...230 V AC 50/60 Hz (AC-3e) 15 kW at 380...400 V AC 50/60 Hz (AC-3e) 15 kW at 415...440 V AC 50/60 Hz (AC-3e) 18.5 kW at 500 V AC 50/60 Hz (AC-3e) 18.5 kW at 660...690 V AC 50/60 Hz (AC-3e)
Motor power hp	2 hp at 115 V AC 50/60 Hz for 1 phase motors 5 hp at 230/240 V AC 50/60 Hz for 1 phase motors 7.5 hp at 200/208 V AC 50/60 Hz for 3 phases motors 10 hp at 230/240 V AC 50/60 Hz for 3 phases motors 20 hp at 460/480 V AC 50/60 Hz for 3 phases motors 30 hp at 575/600 V AC 50/60 Hz for 3 phases motors
Compatibility code	LC1D
Pole contact composition	3 NO
Contact compatibility	M2
Protective cover	With
[Ith] conventional free air thermal current	10 A (at 60 °C) for signalling circuit 50 A (at 60 °C) for power circuit
Irms rated making capacity	140 A AC for signalling circuit conforming to IEC 60947-5-1

12. Siemens 3RT2015-1AK61



Control circuit/ Control:	
Type of voltage of the control supply voltage	AC
Control supply voltage at AC	
• at 50 Hz rated value	110 V
• at 60 Hz rated value	120 V
Operating range factor control supply voltage rated value of magnet coil at AC	
• at 50 Hz	0.8 ... 1.1
• at 60 Hz	0.85 ... 1.1
Apparent pick-up power of magnet coil at AC	
• at 50 Hz	27 V·A
• at 60 Hz	24.3 V·A
Inductive power factor with closing power of the coil	
• at 50 Hz	0.8
• at 60 Hz	0.75
Apparent holding power of magnet coil at AC	
• at 50 Hz	4.2 V·A
• at 60 Hz	3.3 V·A
Inductive power factor with the holding power of the coil	
• at 50 Hz	0.25
• at 60 Hz	0.25
Closing delay	
• at AC	9 ... 35 ms
Opening delay	
• at AC	3.5 ... 14 ms
Arcing time	10 ... 15 ms

Size of contactor	S00
Product expansion	
• function module for communication	No
• Auxiliary switch	Yes
Insulation voltage	
• rated value	690 V
Surge voltage resistance rated value	6 kV
maximum permissible voltage for safe isolation	
• between coil and main contacts acc. to EN 60947-1	400 V
Protection class IP	
• on the front	IP20
• of the terminal	IP20
Degree of pollution	3

13. Motor Induksi SIEMENS 1AV3104A



Electrical data									Safe Area										
U	Δ / Y	f	P	P	I	n	M		$\eta^{3)}$			$\cos\varphi^{3)}$			I_A/I_N	M_A/M_N	M_B/M_N	IE-CL	
[V]		[Hz]	[kW]	[hp]	[A]	[1/min]	[Nm]		4/4	3/4	2/4	4/4	3/4	2/4	I_B/I_N	T_B/T_N	T_B/T_N		
DOL duty (S1) - 155(F) to 130(B)																			
400	Δ	50	3.00	-/-	5.70	2910	9.8		87.1	87.7	87.0	0.87	0.82	0.72	9.4	4.4	4.5	IE3	
460	Δ	60	3.45	-/-	5.60	3505	9.4		88.5	88.5	87.5	0.87	0.83	0.73	10.0	4.0	4.8	IE3	
460	Δ	60	3.00	4.00	5.00	3525	8.1		88.5	88.2	86.7	0.85	0.80	0.69	11.2	4.6	5.6	MG1	
IM B3 / IM 1001			FS 100 L		CC032A		IP55	UKCA	IEC/EN 60034			IEC, EN, UL, CSA, NEMA MG1-12			kVA Code: M				
Environmental conditions : -20 °C - +40 °C / 1,000 m									Locked rotor time (hot / cold) : 9 s 12.1 s										
Mechanical data																			
Sound level (SPL / SWL) at 50Hz 60Hz					67 / 79 dB(A) ^{2) 3)}		71 / 83 dB(A) ^{2) 3)}		External earthing terminal					No					
Moment of inertia					0.0041 kg m²					Vibration severity grade					A				
Bearing DE NDE					6306 Z C3		6206 Z C3		Thermal class					F					
bearing lifetime										Duty type					S1				
L _{10mh} F _{red min} for coupling operation 50 60Hz ¹⁾					40000 h		32000 h		Direction of rotation					bidirectional					
Relubrication interval/quantity DE NDE					5 g 5 g 8000 h					Frame material					aluminum				
Lubricants					Unirex N3					Net weight of the motor (IM B3)					25 kg				
Regreasing device					Yes					Coating (paint finish)					Standard paint finish C2				
Grease nipple					M8x1 DIN 71412					Color, paint shade					RAL7030				
Type of bearing					Preloaded bearing DE					Motor protection					(B) 3 PTC thermistors - for tripping (2 terminals)				
Condensate drainage holes					No					Method of cooling					IC411 - self ventilated, surface cooled				
Terminal box																			
Terminal box position					top					Max. cross-sectional area					4 mm²				
Material of terminal box					Aluminium					Cable diameter from ... to ...					11 mm - 21 mm				
Type of terminal box					TB1 F00					Cable entry					2xM32x1,5-1xM16x1,5				
Contact screw thread					M4					Cable gland					3 plugs				

J. Kesimpulan

Dari hasil praktikum yang sudah dilaksanakan maka dapat saya simpulkan bahwasanya:

1. PLC CP1E-E20SDR adalah sebuah kontroler logika program (PLC) yang digunakan dalam berbagai aplikasi otomatisasi industri. Dengan kemampuan pemrosesan yang cepat dan fitur-fitur yang dapat diprogram, PLC ini mampu mengendalikan proses-produksi secara efisien
2. PLC CP1E-E20SDR memiliki beragam pilihan input dan output (I/O) yang dapat disesuaikan dengan kebutuhan sistem. Ini memungkinkan pengguna untuk menghubungkan berbagai perangkat seperti sensor, aktuator, dan peralatan lainnya dengan mudah
3. PLC ini mendukung berbagai bahasa pemrograman, termasuk ladder logic, yang merupakan bahasa yang umum digunakan dalam pemrograman PLC. Dengan demikian, pengguna dapat mengembangkan program kontrol yang kompleks sesuai dengan kebutuhan spesifik aplikasi mereka